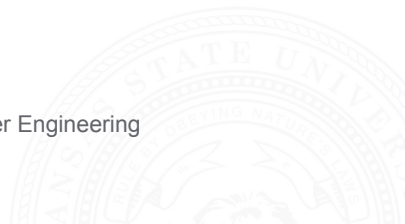


Optimal Fuel Mix Analysis I

Generation utilization, peak adequacy, and ranked demand

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Two parts of today's lecture

Part I - Utilization and adequacy

- capacity factor describes generation over time
- planning reserve margin sets a capacity target above forecast peak
- only the reserve-margin example is a planning application

Part II - Load-duration curves

- reorder a demand profile from highest to lowest
- interpret daily and annual duration axes
- preserve peak, duration, and total energy

PART I

Generation Utilization and Capacity Adequacy

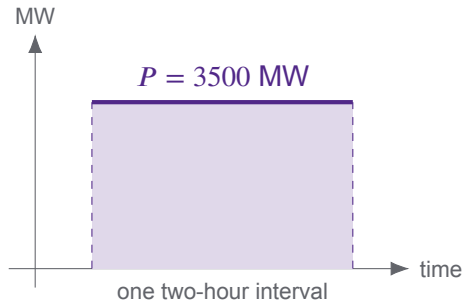
Capacity factor describes generator performance over time
Planning reserve margin supports resource-adequacy planning

Which ratio sets the planning-capacity target?

Demand is a power rate

$$P_{\text{demand}}(t) \quad [\text{MW}]$$

- System demand is the rate at which electrical energy is being used.
- Each demand-profile value is an instant or interval-average power.
- A peak demand is the largest value in the selected record.



Demand-profile height shows power in MW at each point in time.

Power over time gives energy use

$$E_{\text{use}} = \int_{t_1}^{t_2} P_{\text{demand}}(t) dt$$

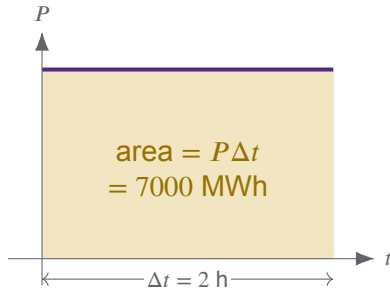
For a constant block,

$$E = P \Delta t.$$

Example

$$3500 \text{ MW} \times 2 \text{ h} = 7000 \text{ MWh.}$$

The rectangle's area is energy.

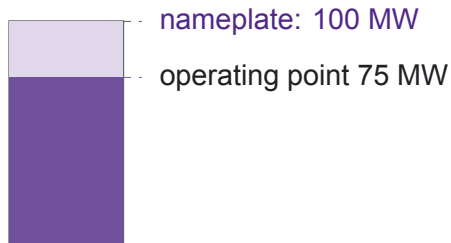


Integrating demand across time converts a power rate in MW into energy use in MWh.

Capacity is generation capability

$C_{\text{gen,nameplate}}$ [MW]

- Generation capacity is a supply-side power capability, not electricity demand.
- Nameplate capacity is the generator's rated maximum output under specified conditions.
- Actual output may be lower; installed generation capacity sums generator nameplates.



Demand describes need, while generation capacity describes available supply. Both use MW.

Rule: do not mix energy and power

Generation utilization

How much energy did the unit generate over a period?

- actual generation: MWh
- rated energy potential over the same period: MWh
- metric introduced next: capacity factor

Capacity adequacy

How much planning capacity is available above forecast peak?

- accredited planning capacity: MW
- forecast peak demand: MW
- planning metric: reserve margin

Start with the physical question, then select the metric with matching units.

Capacity factor measures generation utilization

$$CF = \frac{E_{\text{actual}}}{C_{\text{rated}}T} = \frac{P_{\text{average}}}{C_{\text{rated}}}$$

Numerator

E_{actual} [MWh]

Actual electrical energy generated during the selected period.

Denominator

$C_{\text{rated}}T$ [MWh]

Maximum possible energy if rated power were sustained for the full period.

Capacity factor measures actual generation against the rated energy potential.

Capacity factor example - interpret utilization

1 Maximum annual energy

$$(100 \text{ MW})(8760 \text{ h}) = 876,000 \text{ MWh.}$$

2 Capacity factor

$$\text{CF} = \frac{438,000}{876,000} = 0.500 = 50.0\%.$$

For a 100 MW unit, a 50% capacity factor means an average output of 50 MW over the period, regardless of the exact run-time pattern.

Planning reserve margin is the capacity-planning metric

$$\text{PRM} = \frac{C_{\text{planning}} - P_{\text{peak}}}{P_{\text{peak}}}$$

Numerator - capacity surplus

$$C_{\text{planning}} - P_{\text{peak}} \quad [\text{MW}]$$

Accredited or otherwise credited planning capacity available above the forecast peak.

Denominator - forecast peak

$$P_{\text{peak}} \quad [\text{MW}]$$

The peak demand used for the selected planning period.

PRM compares planning capacity with forecast peak demand. Both quantities use MW.

PRM gives a target, not a complete resource plan

Use PRM to

- convert forecast peak into a planning-capacity target
- compare accredited capacity with that target
- report a capacity surplus or shortfall

Then complete the adequacy study

- assign capacity credit to each resource
- test uncertainty in weather, demand, and outages
- select the resource mix and timing

PRM provides the first capacity check. The full plan also requires accreditation and risk analysis.

Planning example - identify the capacity gap

Wildcatland forecasts a peak demand of 4380 MW, has 4850 MW of accredited planning capacity, and uses a 15% target PRM. Does the system meet the target, and what is the capacity gap?

Compare accredited planning capacity with the PRM-based target, not with annual generation.

First check the current reserve margin

1 Find capacity above peak

$$4850 - 4380 = 470 \text{ MW.}$$

2 Divide by forecast peak

$$\text{PRM}_{\text{current}} = \frac{470}{4380} = 0.107 = 10.7\%.$$

forecast peak

4380 MW

accredited capacity

4850 MW

470 MW
above peak

10.7% < 15% **target**

The system has positive reserve, but it does not meet the 15% target.

Then calculate the capacity needed to reach 15%

1 Calculate target capacity

$$C_{\text{target}} = (1 + 0.15)(4380) = 5037 \text{ MW.}$$

2 Calculate the remaining gap

$$C_{\text{gap}} = 5037 - 4850 = 187 \text{ MW.}$$

forecast peak

4380 MW

accredited capacity

4850 MW

PRM target

5037 MW

187 MW
gap

The plan needs 187 MW of additional accredited capacity.

PRM is a primary metric for U.S. resource-adequacy planning

Capacity factor: performance evaluation

$$CF = \frac{E_{\text{actual}}}{C_{\text{rated}}T}$$

- energy-based productivity measure for a unit or fleet
- supports evaluation of utilization and operating performance

PRM: resource-adequacy planning

$$PRM = \frac{C_{\text{planning}} - P_{\text{peak}}}{P_{\text{peak}}}$$

- compares planning capacity with forecast peak demand
- checked against an area-specific reference level. Full studies add accreditation and probabilistic risk

PRM is a primary capacity metric for U.S. resource-adequacy planning. Capacity factor supports generation-performance evaluation.

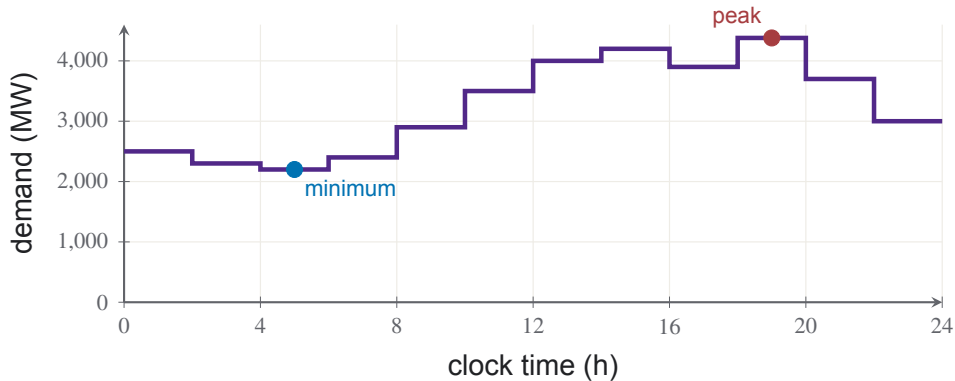
PART II

Load-Duration Curves

Reorder demand by duration while preserving the underlying record

What remains after chronological order is removed?

Chronological demand



Clock order reveals when the peak occurs and how sharply demand changes between intervals.

Load-duration curve

Chronological order

[3500, 2200, 4380, 3000]

- indexed by clock time;
- retains ramps and sequence.

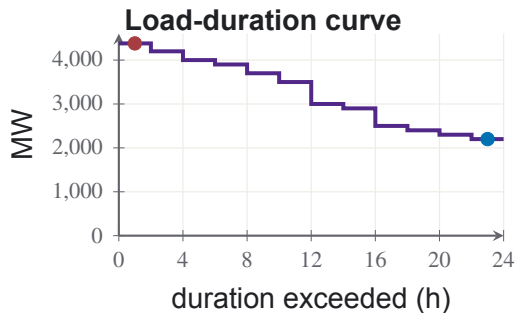
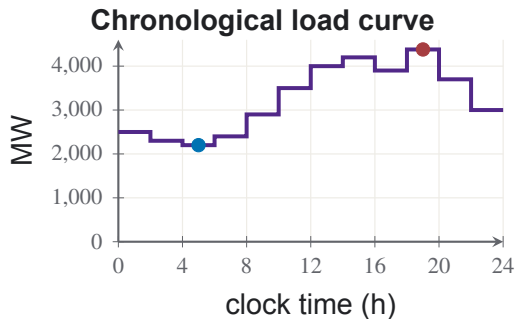
Descending rank

[4380, 3500, 3000, 2200]

- indexed by duration exceeded;
- retains values and time count.

Sorting keeps the demand values and total duration unchanged. Only the horizontal meaning changes.

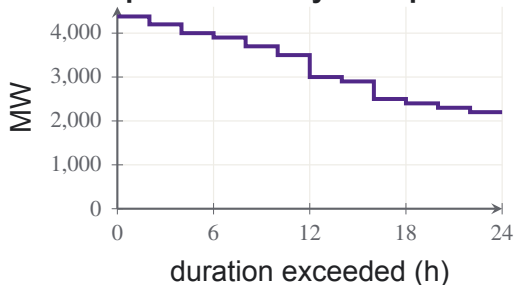
Same demand, new order



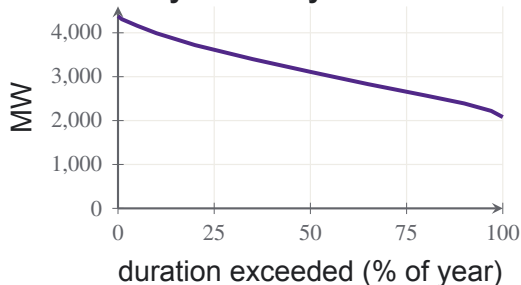
The LDC keeps peak, minimum, duration, and area but loses clock time and ramp information.

Daily and annual LDCs use different source records

Simplified one-day example



Full-year hourly record



The daily example sorts one day's intervals. The annual LDC separately sorts all 8,760 hourly observations.

Construct an annual LDC from hourly demand

Start with the full year

For a non-leap year, use one chronological demand value for every hour:

$$\{P_1, P_2, \dots, P_{8760}\}$$

- consecutive values may come from different daily patterns
- each observation represents one hour

Annual duration comes from counting hourly observations, not from multiplying one day by 365.

Sort and assign duration

$$P_{(1)} \geq P_{(2)} \geq \dots \geq P_{(8760)}$$

- the first k ranked values represent k hours
- annual duration: $d_k = k$ h
- percentage: $100k/8,760$

LDC construction

- 1 **Validate the source profile:** confirm interval length, units, missing values, and covered time.
- 2 **Sort demand in descending order:** retain every value; change only its position.
- 3 **Assign duration:** for equal blocks, each rank adds the same Δt .
- 4 **Verify invariants:** peak, minimum, duration, and area must match the chronological record.

The chronological profile remains the source data. The LDC is a derived view.

SECTION

Worked example daily load profile

Build a 12-block LDC, verify its energy, and calculate the capacity target.



Worked-example data

Block	Clock interval (h)	Demand (MW)
1	0–2	2500
2	2–4	2300
3	4–6	2200
4	6–8	2400
5	8–10	2900
6	10–12	3500

Block	Clock interval (h)	Demand (MW)
7	12–14	4000
8	14–16	4200
9	16–18	3900
10	18–20	4380
11	20–22	3700
12	22–24	3000

Every value represents $\Delta t = 2$ h, so sorting does not require duration weighting.

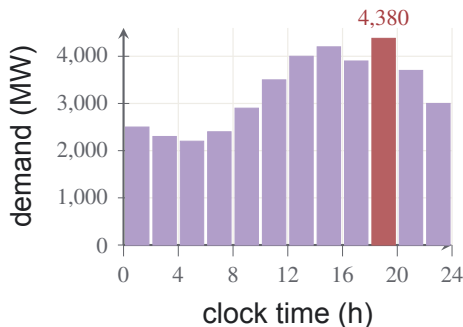
LDC example - Step 1

1 Scan the 24-hour record

The maximum occurs during 18–20 h.

2 Record the peak

$$P_{\text{peak}} = \max(P_1, \dots, P_{12}) = 4380 \text{ MW.}$$



The peak demand is 4380 MW during 18–20 h.

LDC example - Step 2

Ranks 1–6

Rank	Demand (MW)	Clock interval (h)
1	4380	18–20
2	4200	14–16
3	4000	12–14
4	3900	16–18
5	3700	20–22
6	3500	10–12

Ranks 7–12

Rank	Demand (MW)	Clock interval (h)
7	3000	22–24
8	2900	8–10
9	2500	0–2
10	2400	6–8
11	2300	2–4
12	2200	4–6

Sorting changes the order, but each demand value keeps its original time interval.

LDC example - Step 3

For rank k with $\Delta t = 2$ h,

$$x_k = (k - 1)\Delta t.$$

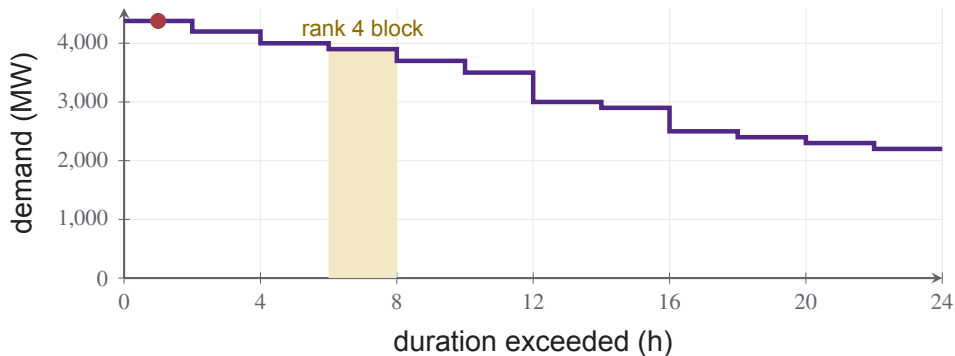
The rank- k demand occupies

$$x_k \leq x < x_k + \Delta t.$$

Rank	Duration (h)	Demand (MW)
1	0–2	4380
4	6–8	3900
7	12–14	3000
12	22–24	2200

The horizontal coordinate now shows hours exceeded.

LDC example - Step 4



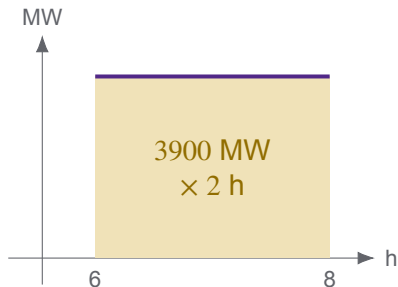
Every rectangle contributes to energy, and the first step sets the peak.

One LDC block: energy

Rank 4 occupies duration 6–8 h at 3900 MW.

$$\Delta E_4 = P_4 \Delta t$$

$$\Delta E_4 = (3900 \text{ MW})(2 \text{ h}) = \boxed{7800 \text{ MWh}}.$$



An LDC height is power. A block's area is energy.

Daily-energy check

$$E_{\text{day}} = \sum_{k=1}^{12} P_k(2 \text{ h}) = (38,980 \text{ MW})(2 \text{ h}) = 77,960 \text{ MWh.}$$

Chronological area

77,960 MWh

Original clock order.

LDC area

77,960 MWh

Descending rank order.

Sorting changes sequence, so the energy check must agree exactly.

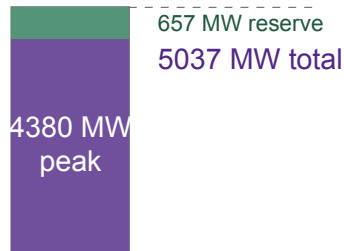
From LDC peak to capacity target

Inputs preserved and defined in L03

$$P_{\text{peak}} = 4380 \text{ MW}, \quad \text{PRM} = 15\%.$$

$$\begin{aligned} C_{\text{req}} &= (1 + \text{PRM})P_{\text{peak}} \\ &= 1.15(4380) = \boxed{5037 \text{ MW}}. \end{aligned}$$

The peak is unchanged when the demand values are sorted.



The LDC preserves peak demand, and PRM converts it into a capacity target.

A brief link to the Final Project

One result from L03 carries forward

$$P_{\text{peak}} = 4380 \text{ MW}, \quad \text{PRM} = 15\% \quad \implies \quad C_{\text{target}} = 5037 \text{ MW}.$$

- We will use this value as the initial planning-capacity target in the Final Project.
- The generation-mix decision begins in L04.

For now, carry the 5037 MW capacity target forward.

Summary

Capacity factor

$$\frac{E_{\text{actual}}}{C_{\text{rated}}T}$$

Evaluates generation use over a period.

Planning reserve margin

$$\frac{C_{\text{planning}} - P_{\text{peak}}}{P_{\text{peak}}}$$

Measures the capacity cushion above peak demand.

Load-duration curve

$$P_{\text{LDC}}(\tau)$$

Shows how long each demand level is met or exceeded.

PRM sets the peak-capacity target; the annual LDC adds demand-duration information.